

FGFR2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR2 gene fusions from the msk_impact_50k_2026 study, covering 136 fusion events. 108/136 fusions (79.4%) were in-frame. 115/136 fusions (84.6%) retained the Protein tyrosine and serine/threonine kinase domain. Domain retention was tested with Fisher's exact test ($p=0.000424687$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

Domain retention

Protein tyrosine and serine/threonine kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain, $p=0.000424687$ (statistically significant at $\alpha=0.05$). The Protein tyrosine and serine/threonine kinase appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Domain disruption

Disruption of the Immunoglobulin domain and Immunoglobulin I-set domain was tested with Fisher's exact test comparing in-frame fusions against all others; 9/108 (8.3%) of in-frame fusions disrupted the domain, $p=0.989695$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

Cutpoint detection

Cutpoint detection scanned 134 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 732 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -25 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=115$) and "not_retained" ($n=19$). For Frame_status, retained: 99/102 (97.1%, 95% CI 91.7%-99.0%); not_retained: 9/14 (64.3%, 95% CI 38.8%-83.7%). Welch's t-test comparing Tumor_variant_count between groups gave means of 104.965 vs 61.8947, $p=0.20677$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (78 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
BICC1	28	0.20588235294117646	0.20043213737826426	0.20043213737826426	0.9354064039408867	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.36618079938289244	1
TACC2	8	0.058823529411764705	0.20043213737826426	0.20043213737826426	0.9314655172413793	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3214720193191428	2
ANKRD11	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	1.0	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3163110152623477	3
NRG3	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9698275862068966	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31399103554632335	4
NOL4	4	0.029411764705882353	0.20043213737826426	0.20043213737826426	0.9392241379310344	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31381228301082637	5
KIAA1217	4	0.029411764705882353	0.20043213737826426	0.20043213737826426	0.9387931034482758	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3137476278384126	6
TBC1D4	3	0.022058823529411766	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.311671055830299	7
CTNNA3	3	0.022058823529411766	0.20043213737826426	0.20043213737826426	0.9390804597701149	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31158484893374727	8
SHTN1	3	0.022058823529411766	0.20043213737826426	0.20043213737826426	0.9385057471264368	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31149864203719557	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
INA	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30946517347735786	10
SORBS1	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30946517347735786	11
TTC28	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30946517347735786	12
WAC	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30946517347735786	13
AHCYL1	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9387931034482758	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30933586313253025	14
BAIAP2L1	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9387931034482758	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30933586313253025	15
NRBF2	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30920655278770265	16
OPTN	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30920655278770265	17
ABLM1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	18
ADAM9	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AMOT	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	20
ATAD1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	21
ATP6V1D	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	22
BICD1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	23
BNIP2	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	24
CAT	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	25
CCDC6	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	26
CEACAM7	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	27
CEP112	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	28
DDX21	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DIABLO	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	30
GAB2	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	31
HSPA12A	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	32
KLHL29	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	33
MARVELD3	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	34
MYO18B	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	35
NRAP	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	36
PDZRN4	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	37
PKP4	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	38
POC1B	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRDX3	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	40
RNF145	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	41
RPAP3	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	42
SH3GLB1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	43
SHC2	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	44
SKI	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	45
SRRT	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	46
TFEC	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	47
VCL	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	48
VPS26A	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
YTHDF3	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	50
ZMYM4	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9396551724137931	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30725929112441663	51
BTBD16	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	52
CCDC186	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	53
CCDC73	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	54
KIF14	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	55
L3MBTL3	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	56
LAMC1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	57
MBIP	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	58
MYH15	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MYH9	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	60
MYLK	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	61
NID2	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	62
PAH	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	63
PALLD	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	64
PPP1R21	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	65
RABGAP1L	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	66
ROCK1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	67
SEPTIN10	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	68
STAT4	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	69

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
UBP1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	70
ZFAT	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.9379310344827586	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3070006704347615	71
FOXP1	2	0.014705882352941176	0.20043213737826426	0.20043213737826426	0.9103448275862069	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3050686217532199	72
ADAM12	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.903448275862069	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.301828256641658	73
KCNH7	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.8862068965517241	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2992420497451063	74
ATE1	3	0.022058823529411766	0.20043213737826426	0.20043213737826426	0.8080459770114943	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.29192967651995416	75
PRKAG2	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	None	0.739106710891876	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.19566001795570317	76
ST8SIA6	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	None	0.739106710891876	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.19566001795570317	77
GRID1	1	0.007352941176470588	0.20043213737826426	0.20043213737826426	0.0	0.739106710891876	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1663110152623477	78

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
152	0	115	1	18	0.0	0.1417910447761194	0.8483511974119787

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
469	0	115	2	17	0.0	0.019189765458422176	1.7169303332757584
676	0	115	3	16	0.0	0.0024714091878270984	2.607055343103334
732	0	115	5	14	0.0	3.4829019675778254e-05	4.458058749354329
767	63	52	15	4	0.3230769230769231	0.07677359623872276	1.1147881156383952
768	111	4	19	0	0.0	1.0	-0.0
798	112	3	19	0	0.0	1.0	-0.0
801	113	2	19	0	0.0	1.0	-0.0
802	114	1	19	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

9	6
99	20

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf13895	Immunoglobulin domain	134	119	0	15	0.0
auto_disruption_pf07679	Immunoglobulin I-set domain	134	118	0	16	0.0

domain_retention tables

frame_domain_contingency_table

99	16
9	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	134	115	3	16	0.09006270188105643

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	117	136	0.8602941176470589

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABLM1	1
ADAM12	1
ADAM9	1
AHCYL1	2
AMOT	1
ANKRD11	1
ATAD1	1
ATE1	3
ATP6V1D	1
BAIAP2L1	2
BICC1	28
BICD1	1
BNIP2	1
BTBD16	1
CAT	1
CCDC186	1
CCDC6	1
CCDC73	1
CEACAM7	1
CEP112	1
CTNNA3	3
DDX21	1
DIABLO	1
FOXP1	2
GAB2	1
GRID1	1
HSPA12A	1
INA	2

Partner_gene	Event_count
KCNH7	1
KIAA1217	4
KIF14	1
KLHL29	1
L3MBTL3	1
LAMC1	1
MARVELD3	1
MBIP	1
MYH15	1
MYH9	1
MYLK	1
MYO18B	1
NID2	1
NOL4	4
NRAP	1
NRBF2	2
NRG3	2
OPTN	2
PAH	1
PALLD	1
PDZRN4	1
PKP4	1
POC1B	1
PPP1R21	1
PRDX3	1
PRKAG2	1
RABGAP1L	1
RNF145	1
ROCK1	1
RPAP3	1
SEPTIN10	1
SH3GLB1	1
SHC2	1
SHTN1	3
SKI	1

Partner_gene	Event_count
SORBS1	2
SRRT	1
ST8SIA6	1
STAT4	1
TACC2	8
TBC1D4	3
TFEC	1
TTC28	2
UBP1	1
VCL	1
VPS26A	1
WAC	2
YTHDF3	1
ZFAT	1
ZMYM4	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021826-T01-IM6-1	P-0021826-T01-IM6		ANKRD11-FGFR2	ANKRD11	unknown	five_prime	123243799	17	732	True	disrupted	0.9097472924187726	True	Immunoglobulin domain; Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10;16)(q26.13;q24.3)(chr10:g.123243799::chr16:g.89538175) Transcript Fusion {FGFR2:ANKRD11}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013241-T01-IM5-2	P-0013241-T01-IM5		ATP6V1D-FGFR2	ATP6V1D	in-frame	five_prime	123242048	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - ATP6V1D (NM_015994) rearrangement: t(10;14)(q26.13;q23.2)(chr10:g.123242048::chr14:g.67818116) Protein fusion: in frame (FGFR2-ATP6V1D)	NA
EVT-P-0009512-T01-IM5-3	P-0009512-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123240449	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			Note: The FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal rearrangement event is an inversion which results in the fusion of FGFR2 exons 1-17 with BICC1 exons 3-21 . The fusion is predicted to be in frame and includes the kinase domain of FGFR2. Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0002675-T01-IM3-4	P-0002675-T01-IM3		BICC1-FGFR2	BICC1	in-frame	five_prime	123241096	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			NA Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 9067-T01-IM6-5	P-0029 067-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123240197	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21) : c.2302-67:FGFR2_c.238-6757:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-000 3177-T01-IM5-6	P-0003 177-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123242188	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			NA Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-002 5807-T01-IM6-7	P-0025 807-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123242181	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusion_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_p_o_int_exo_n	breakpoint_protein_p_osition	is_intro_nic_bre_akpoint	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_ncated	retain_ed_do_mains	lost_dom_ains	disrupt_ed_do_mains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-002 5250-T01-I M6-8	P-0025 250-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	thr_ee_prime	123240 491	18	768	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 16-21): c.2302-956:FGFR2_ c.2182-297:BICC1inv Protein Fusion: in frame {BICC1:FGFR2}	NA
EVT-P-003 2033-T01-I M6-9	P-0032 033-T0 1-IM6		BICC1- FGFR2	BIC C1	in-fr ame	fiv_e_prime	123241 685	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c. 2301+1527:FGFR2_c.237+3 8877:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-000 5538-T01-I M5-10	P-0005 538-T0 1-IM5		BICC1- FGFR2	BIC C1	in-fr ame	fiv_e_prime	123242 753	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:F GFR2_c.2182-276:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020186-T01-IM6-11	P-0020186-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241148	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exon17 fused with BICC1 exon3): c.2302-1613:FGFR2_c.237+32254: BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0020025-T01-IM6-12	P-0020025-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241984	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 3-21) : c.2301+1228:FGFR2_c.238-4773:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0018115-T01-IM6-13	P-0018115-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241970	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused in-frame with BICC1 exons 3-21): c.2301+1242:FGFR2_c.237+34947:BICC1dup Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015942-T01-IM6-14	P-0015942-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241020	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exons 1-17 with BICC1 exons 9-21) : c.2302-1485:FGFR2_c.1048-60:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-0008146-T01-IM5-15	P-0008146-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123240154	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2302-619:FGFR2_c.2182-744:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-0037617-T01-IM6-16	P-0037617-T01-IM6		BICC1-FGFR2	BICC1	in-frame	three_prime	123241302	18	768	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain		BICC1 (NM_001080512) - FGFR2 (NM_000141) rearrangement: c.238-34060:BICC1_c.2302-1767:FGFR2inv Protein Fusion: in frame {BICC1:FGFR2}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 5432-T01-I M6-17	P-0015 432-T0 1-IM6		CAT-F GFR2	CAT	in-fr ame	fiv e_ pri me	123242 534	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26. 13;p13)(chr10:g.123242534:: chr11:g.34467980) Protein fusion: in frame (FGFR2-CAT)	NA
EVT-P-002 7305-T01-I M6-18	P-0027 305-T0 1-IM6		CCDC 6-FGF R2	CC DC6	in-fr ame	fiv e_ pri me	123241 542	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+ 1670:FGFR2_c.582+6426:C CDC6del Protein Fusion: in frame {FGFR2:CCDC6}	NA
EVT-P-002 7741-T01-I M6-19	P-0027 741-T0 1-IM6		CEAC AM7-F GFR2	CEA CA M7	unk now n	fiv e_ pri me	123241 990	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) rearrangement: t(10;19)(q26. 13;q13.2)(chr10:g.12324199 0::chr19:g.42175729) Transcript Fusion {FGFR2:CEACAM7}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 3724-T01-I M5-20	P-0003 724-T0 1-IM5		CEP11 2-FGF R2	CEP 112	in-fr ame	thre e_ pr ime	123241 923	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eonine ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		t(17;10)(q24.1;q26.13)(chr17:g.64023428::chr10:g.123241923) Protein fusion: in frame (CCDC46-FGFR2)	NA
EVT-P-001 9723-T01-I M6-22	P-0019 723-T0 1-IM6		CTNN A3-FG FR2	CTN NA3	in-fr ame	fiv e_ pr ime	123243 018	17	767	True	retain ed	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194: FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 1516-T01-IM5-23	P-0011 516-T01-IM5		CTNNA3-FGFR2	CTNNA3	in-frame	threonine_prime	123241940	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain		CTNNA3 (NM_013266) -FGFR2 (NM_000141) Rearrangement : c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)	NA
EVT-P-002 0410-T01-IM6-24	P-0020 410-T01-IM6		DDX21-FGFR2	DDX21	in-frame	five_prime	123243048	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - DDX21 (NM_004728) fusion (FGFR2 exons 1-17 fused to DDX21 exons 2-15): c.2301+164:FGFR2_c.88-860:DDX21inv Protein Fusion: in frame {FGFR2:DDX21}	NA
EVT-P-001 8869-T03-IM6-26	P-0018 869-T03-IM6		FGFR2-ABLIM1	ABLIM1	in-frame	five_prime	123242601	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - ABLIM1 (NM_002313) fusion: c.2301+611:FGFR2_c.895-12244:ABLIM1del Protein Fusion: in frame {FGFR2:ABLIM1}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-002 6275-T01-I M6-27	P-0026 275-T0 1-IM6		FGFR2 -ADAM 12	ADA M12	unk now n	fiv e_ pri me	123246 899	15	676	False	disru pted	0.7075812 27436823 1	True	Immun oglobu lin do main; I mmun oglobu lin t-set d omain		Protein tyrosin e and s erine/th reonine kinase	FGFR2 (NM_000141) - ADAM12 (NM_003474) rearrangement: c.2026:FGF R2_c.1618+5260:ADAM12d up Protein Fusion: mid-exon {FGFR2:ADAM12}	NA
EVT-P-002 0164-T01-I M6-28	P-0020 164-T0 1-IM6		FGFR2 -ADAM 9	ADA M9	in-fr ame	fiv e_ pri me	123242 738	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin t-set d omain			FGFR2 (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10(p11.22;q26.13) (chr8:g.38901161::chr10:g.1 23242738) Protein Fusion: in frame {FGFR2:ADAM9}	NA
EVT-P-001 8920-T01-I M6-29	P-0018 920-T0 1-IM6		FGFR2 -AHCY L1	AH CYL 1	in-fr ame	fiv e_ pri me	123242 706	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin t-set d omain			FGFR2 (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AHCYL1 exons 2-17): t(1;10)(p13.3;q2 6.13)(chr1:g.110531185::chr 10:g.123242706) Protein Fusion: in frame {FGFR2:AHCYL1}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 0644-T01-I M3-30	P-0000 644-T0 1-IM3		FGFR2 -AHCY L1	AH CYL 1	in-fr ame	thre _pr ime	123239 745	18	768	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		NA Protein fusion: in frame (AHCYL1-FGFR2)	NA
EVT-P-001 2666-T01-I M5-31	P-0012 666-T0 1-IM5		FGFR2 -AMOT	AM OT	in-fr ame	fiv e_ pri me	123241 762	17	767	True	retai ned	1.0	False	Protei n tyro sine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - AMOT (NM_001113490) Rearrangement : t{(10;X)(q26 .13.1;q23)(chr10:g.12324176 2::chrX:g.112058354) Protein fusion: in frame (FGFR2-AMOT)	NA
EVT-P-004 7504-T01-I M6-32	P-0047 504-T0 1-IM6		FGFR2 -ATAD 1	ATA D1	in-fr ame	fiv e_ pri me	123241 680	17	767	True	retai ned	1.0	False	Protei n tyro sine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - ATAD1 (NM_032810) fusion: c.2301+1532:FGFR2_c.831+ 3993:ATAD1del Protein Fusion: in frame {FGFR2:ATAD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 7424-T01-I M6-33	P-0017 424-T0 1-IM6		FGFR2 -ATE1	ATE 1	out-of-frame	five_prime	123240 443	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908: FGFR2_c.1379-520:ATE1dup Protein Fusion: out of frame {FGFR2:ATE1}	NA
EVT-P-006 3512-T01-I M7-35	P-0063 512-T0 1-IM7		FGFR2 -ATE1	ATE 1	unknown	five_prime	123263 336	10	469	False	lost	0.0	False	Immunoglobulin domain; Immunoglobulin I-set domain	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1}	NA
EVT-P-006 3625-T01-I M7-36	P-0063 625-T0 1-IM7		FGFR2 -ATE1	ATE 1	in-frame	five_prime	123242 847	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 2794-T01-I M6-38	P-0052 794-T0 1-IM6		FGFR2 -BAIAP 2L1	BAI AP2 L1	in-fr ame	fiv e_ pri me	123240 813	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::chr10:g.12 3240813) Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA
EVT-P-005 9296-T01-I M7-39	P-0059 296-T0 1-IM7		FGFR2 -BAIAP 2L1	BAI AP2 L1	in-fr ame	fiv e_ pri me	123242 027	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::chr10:g.12 3242027) Protein Fusion: in frame {FGFR2:BAIAP2L1}	NA
EVT-P-000 4520-T01-I M5-41	P-0004 520-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 269	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2(NM_000141) - BICC1(NM_001080512) Fusion (FGFR2 exon 17 fused with exon 10 of BICC1) : c.2302-734:FGFR2_c.1180 -924:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 3174-T01-I M5-42	P-0013 174-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 123	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c. 2302-588:FGFR2_c.2182-28 4:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA
EVT-P-004 9135-T07-I M7-43	P-0049 135-T0 7-IM7		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123239 791	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c .238-36290:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-006 8640-T01-I M7-44	P-0068 640-T0 1-IM7		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 444	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-909:FGFR2_c .238-18504:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-004 9135-T03-I M6-45	P-0049 135-T0 3-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123239 791	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-256:FGFR2_c .238-36290:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-004 9133-T04-I M6-46	P-0049 133-T0 4-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123242 098	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2 _c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-000 6885-T01-I M5-48	P-0006 885-T0 1-IM5		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123240 587	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) rearrangement: c.2302-1052: FGFR2_c.238-5239:BICC1in v Protein fusion: in frame (FGFR2-BICC1)	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 1773-T01-I M6-49	P-0051 773-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123242 828	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+384:FGFR2_ c.237+13961:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-004 9133-T02-I M6-50	P-0049 133-T0 2-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123242 098	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+384:FGFR2_ c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-005 3822-T01-I M6-51	P-0053 822-T0 1-IM6		FGFR2 -BICC1	BIC C1	in-fr ame	fiv e_ pri me	123243 033	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+179:FGFR2_ c.238-1932:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 1327-T01-I M6-52	P-0021 327-T0 1-IM6		FGFR2 -BICC1	BICC1	in-frame	five_prime	123240 344	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-1613:FGFR2_c.237+32254:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-006 0245-T01-I M7-53	P-0060 245-T0 1-IM7		FGFR2 -BICC1	BICC1	unknown	five_prime	123239 377	18	820	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.1179+1103:BICC1inv Protein Fusion: mid-exon {FGFR2:BICC1}	NA
EVT-P-006 8033-T01-I M7-54	P-0068 033-T0 1-IM7		FGFR2 -BICC1	BICC1	in-frame	five_prime	123239 673	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c.238-40422:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 9087-T01-IM6-55	P-0039 087-T01-IM6		FGFR2-BICC1	BICC1	in-frame	five_prime	123242059	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21) : c.2301+1153:FGFR2_c.2181+133:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA
EVT-P-005 4191-T01-IM6-56	P-0054 191-T01-IM6		FGFR2-BICD1	BICD1	in-frame	five_prime	123242849	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BICD1 (NM_001714) fusion: t(10;12)(q26.13;p11.21)(chr10:g.123242849::chr12:g.32465127) Protein Fusion: in frame {FGFR2:BICD1}	NA
EVT-P-006 1764-T01-IM7-57	P-0061 764-T01-IM7		FGFR2-BNIP2	BNIP2	in-frame	five_prime	123243170	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BNIP2 (NM_004330) fusion: t(10;15)(q26.13;q22.2)(chr10:g.123243170::chr15:g.59975666) Protein Fusion: in frame {FGFR2:BNIP2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 0135-T01-IM6-58	P-0020 135-T01-IM6		FGFR2-BTBD16	BTBD16	out-of-frame	five_prime	123239702	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - BTBD16 (NM_144587) rearrangement: c.2302-167: FGFR2_c.386-107: BTBD16inv Protein Fusion: out of frame {FGFR2:BTBD16}	NA
EVT-P-003 3652-T01-IM6-59	P-0033 652-T01-IM6		FGFR2-CCDC186	CCDC186	in-frame	five_prime	123240830	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - CCDC186(NM_018017) Rearrangement : c.2302-1295:FGFR2_c.1222-3356:CCDC186del Protein Fusion: in frame {FGFR2:CCDC186}	NA
EVT-P-006 0582-T01-IM7-60	P-0060 582-T01-IM7		FGFR2-CCDC73	CCDC73	in-frame	five_prime	123241248	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - CCDC73 (NM_001008391) fusion: t(10;11)(q26.13;p13)(chr10:g.123241248::chr11:g.32747346) Protein Fusion: in frame {FGFR2:CCDC73}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 7574-T01-IM6-61	P-0037 574-T01-IM6		FGFR2-CTNNA3	CTNNA3	in-frame	five_prime	123240911	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - CTNNA3 (NM_013266) fusion: c.2302-1376:FGFR2_c.1884+30653:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3}	NA
EVT-P-004 4768-T01-IM6-63	P-0044 768-T01-IM6		FGFR2-DIABLO	DIABLO	in-frame	five_prime	123243152	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - DIABLO (NM_019887) fusion: t(10;12)(q26.13;q24.31)(chr10:g.123243152::chr12:g.122708063) Protein Fusion: in frame {FGFR2:DIABLO}	NA
EVT-P-005 8539-T01-IM6-101	P-0058 539-T01-IM6		FGFR2-FOXP1	FOXP1	unknown	five_prime	123239434	18	801	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71113500::chr10:g.123239434) Protein Fusion: mid-exon {FGFR2:FOXP1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006 1783-T01-I M7-102	P-0061 783-T01-IM7		FGFR2-FOXP1	FOX P1	in-frame	five_prime	123241483	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - FOXP1 (NM_001244814) fusion: t(3;10)(p13;q26.13)(chr3:g.71192573::chr10:g.123241483) Protein Fusion: in frame {FGFR2:FOXP1}	NA
EVT-P-005 5215-T01-I M6-103	P-0055 215-T01-IM6		FGFR2-GRID1	GRID1	out-of-frame	five_prime	123323934	4	152	True	lost	0.0	False	Immunoglobulin domain	Protein tyrosine and serine/threonine kinase ; Immunoglobulin I-set domain		FGFR2 (NM_000141) - GRID1 (NM_017551) rearrangement: c.454+82:FGFR2_c.726+103661:GRID1del Protein Fusion: out of frame {FGFR2:GRID1}	NA
EVT-P-001 5437-T01-I M6-104	P-0015 437-T01-IM6		FGFR2-HSPA12A	HSPA12A	in-frame	five_prime	123242591	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - HSPA12A (NM_025015) rearrangement: c.2301+621:FGFR2_c.663+3346:HSPA12A del Protein fusion: in frame {FGFR2-HSPA12A}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009701-T01-IM5-105	P-0009701-T01-IM5		FGFR2-KCNH7	KCNH7	unknown	five_prime	123239444	18	798	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - KCNH7 (NM_033272) Rearrangement : t(2;10)(q24.3; q26.13) (chr2:g.163263969::chr10:g.123239444) Protein fusion: mid-exon (FGFR2-KCNH7)	NA
EVT-P-0008658-T01-IM5-107	P-0008658-T01-IM5		FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241193	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2302-1658:FGFR2_c.355-22274:KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0034784-T01-IM6-108	P-0034784-T01-IM6		FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123242254	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) fusion: c.2301+958:FGFR2_c.355-49659:KIAA1217inv Protein Fusion: in frame {FGFR2:KIAA1217}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013405-T01-IM5-109	P-0013405-T01-IM5		FGFR2-KIAA1217	KIAA1217	in-frame	five_prime	123241766	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - KIAA1217 (NM_019590) Rearrangement: c.2301+1446:FGFR2_c.354+73179:KIAA1217inv Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-0003660-T01-IM5-110	P-0003660-T01-IM5		FGFR2-KIF14	KIF14	in-frame	five_prime	123240038	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			t(1;10)(q32.1;q26.13)(chr1:g.200555931::chr10:g.123240038) Protein fusion: in frame (FGFR2-KIF14)	NA
EVT-P-0013241-T01-IM5-111	P-0013241-T01-IM5		FGFR2-KLHL29	KLHL29	unknown	three_prime	123241923	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain		KLHL29 (NM_052920) - FGFR2 (NM_000141) rearrangement: t(2;10)(p24.1;q26.13)(chr2:g.23647979::chr10:g.123241923) Transcript fusion (KLHL29-FGFR2)	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 9439-T02-I M6-112	P-0019 439-T0 2-IM6		FGFR2 -L3MB TL3	L3M BTL 3	in-fr ame	fiv e_ pri me	123239 680	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - L3MBTL3 (NM_032438) rearrangement: t(6;10)(q23.1 ;q26.1)(chr6:g.130408449::c hr10:g.123239680) Protein Fusion: in frame {FGFR2:L3MBTL3}	NA
EVT-P-001 1394-T01-I M5-113	P-0011 394-T0 1-IM5		FGFR2 -LAMC 1	LAM C1	in-fr ame	fiv e_ pri me	123240 042	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - LAMC1 (NM_002293) rearrangement: t(1;10)(q25.3; q26.13)(chr1:g.183109471::c hr10:g.123240042) Protein fusion: in frame (FGFR2-LAMC1)	NA
EVT-P-006 4855-T01-I M7-115	P-0064 855-T0 1-IM7		FGFR2 -MBIP	MBI P	in-fr ame	fiv e_ pri me	123240 070	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - MBIP (NM_016586) fusion: t(10;14) (q26.13;q13.3)(chr10:g.123 240070::chr14:g.36787130) Protein Fusion: in frame {FGFR2:MBIP}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 0844-T01-I M6-116	P-0030 844-T0 1-IM6		FGFR2 -MYH1 5	MY H15	in-fr ame	fiv e_ pri me	123240 665	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.1 3;q26.13)(chr3:g.108134575: :chr10:g.123240665) Protein Fusion: in frame {FGFR2:MYH15}	NA
EVT-P-002 7623-T01-I M6-117	P-0027 623-T0 1-IM6		FGFR2 -MYLK	MYL K	unk now n	fiv e_ pri me	123240 959	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1 ;q26.13)(chr3:g.123525612:: chr10:g.123240959) Transcript Fusion {FGFR2:MYLK}	NA
EVT-P-003 8235-T01-I M6-118	P-0038 235-T0 1-IM6		FGFR2 -NOL4	NOL 4	in-fr ame	fiv e_ pri me	123240 418	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t (10;18)(q26.13;q12.1)(chr10: g.123240418::chr18:g.31580 817) Protein Fusion: in frame {FGFR2:NOL4}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 2423-T02-I M6-119	P-0012 423-T0 2-IM6		FGFR2 -NOL4	NOL 4	in-fr ame	fiv e_ pri me	123242 055	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1) (chr10:g.123242055::chr18:g .31563418) Protein Fusion: in frame {FGFR2:NOL4}	NA
EVT-P-000 6328-T02-I M6-120	P-0006 328-T0 2-IM6		FGFR2 -NRBF 2	NR BF2	in-fr ame	fiv e_ pri me	123241 246	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157- 363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2}	NA
EVT-P-005 8339-T01-I M6-121	P-0058 339-T0 1-IM6		FGFR2 -PAH	PAH	in-fr ame	fiv e_ pri me	123240 026	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12) (q26.13;q23.2)(chr10:g.123 240026::chr12:g.103294473) Protein Fusion: in frame {FGFR2:PAH}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 5500-T01-I M6-122	P-0015 500-T0 1-IM6		FGFR2 -PALL D	PAL LD	in-fr ame	fiv e_ pri me	123239 869	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - PALLD (NM_001166108) Rearrangement : t(4;10)(q33, q26.13)(chr4:g.169605121::c hr10:g.123239869) Protein fusion: in frame (FGFR2-PALLD)	NA
EVT-P-002 1448-T03-I M6-123	P-0021 448-T0 3-IM6		FGFR2 -PKP4	PKP 4	in-fr ame	fiv e_ pri me	123242 661	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - PKP4 (NM_003628) rearrangement: t(2;10)(q24.1 ;q26.13)(chr2:g.159395413:: chr10:g.123242661) Protein Fusion: in frame {FGFR2:PKP4}	NA
EVT-P-005 9669-T01-I M7-125	P-0059 669-T0 1-IM7		FGFR2 -POC1 B	PO C1B	in-fr ame	fiv e_ pri me	123242 798	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - POC1B (NM_001199777.2) fusion: t(10;12)(q26.13;q21.3 3)(chr10:g.123242798::chr12 :g.89857934) Protein Fusion: in frame {FGFR2:POC1B}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008657-T01-IM5-126	P-0008657-T01-IM5		FGFR2-PPP1R21	PPP1R21	in-frame	five_prime	123240684	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR1 (NM_000141) - PPP1R21 (NM_152994) Rearrangement : t(2;10)(2p16.3;10q26.13)(chr2:g.48721864::chr10:g.123240684) Protein fusion: in frame (FGFR2-KLRAQ1)	NA
EVT-P-0057247-T02-IM7-127	P-0057247-T02-IM7		FGFR2-PRDX3	PRDX3	in-frame	five_prime	123242894	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c.2301+318:FGFR2_c.36+287:PRDX3del Protein Fusion: in frame {FGFR2:PRDX3}	NA
EVT-P-0011394-T01-IM5-129	P-0011394-T01-IM5		FGFR2-RABGAP1L	RABGAP1L	in-frame	five_prime	123240065	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.123240065) Protein fusion: in frame (FGFR2-RABGAP1L)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 2639-T01-IM5-130	P-0012 639-T01-IM5		FGFR2-SEPTIN10	SEPTIN10	unknown	five_prime	123240798	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - SEPT10 (NM_144710) rearrangement: t(2;10)(q13;q26.13)(chr2:g.110325395::chr10:g.123240798) Protein fusion: mid-exon (FGFR2-SEPT10)	NA
EVT-P-005 5202-T01-IM6-131	P-0055 202-T01-IM6		FGFR2-SH3GLB1	SH3GLB1	in-frame	five_prime	123242019	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - SH3GLB1 (NM_001206651) rearrangement: t(1;10)(p22.3;q26.13)(chr1:g.87191292::chr10:g.123242019) Protein Fusion: in frame {FGFR2:SH3GLB1}	NA
EVT-P-001 1516-T01-IM5-132	P-0011 516-T01-IM5		FGFR2-SHC2	SHC2	in-frame	five_prime	123243182	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - SHC2 (NM_012435) Rearrangement : t(10;19)(q26.13;p13.3)(chr10:g.123243182::chr19:g.447779) Protein fusion: in frame (FGFR2-SHC2)	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 2152-T01-I M6-133	P-0032 152-T0 1-IM6		FGFR2 -SHTN 1	SHT N1	in-fr ame	fiv e_ pri me	123241 513	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2301+1699:FGFR2 _c.534+751:KIAA1598 Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-005 7005-T01-I M6-134	P-0057 005-T0 1-IM6		FGFR2 -SHTN 1	SHT N1	in-fr ame	fiv e_ pri me	123239 901	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) fusion: c.2302-366:FGFR2_c .534+660:KIAA1598del Protein Fusion: in frame {FGFR2:KIAA1598}	NA
EVT-P-006 4573-T01-I M7-135	P-0064 573-T0 1-IM7		FGFR2 -SKI	SKI	in-fr ame	fiv e_ pri me	123242 232	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - SKI (NM_003036) fusion: t(1;10)(p36.33;q26.13)(chr1:g.22078 11::chr10:g.123242232) Protein Fusion: in frame {FGFR2:SKI}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0054244-T01-IM6-136	P-0054244-T01-IM6		FGFR2-SORBS1	SORBS1	in-frame	five_prime	123243201	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - SORBS1 (NM_001034954) fusion: c.2301+11:FGFR2_c.1738-382:SORBS1del Protein Fusion: in frame {FGFR2:SORBS1}	NA
EVT-P-0057277-T01-IM6-137	P-0057277-T01-IM6		FGFR2-SRRT	SRRT	out-of-frame	five_prime	123242651	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - SRRT (NM_015908) fusion: t(7;10)(q22.1;q26.13)(chr7:g.100475766::chr10:g.123242651) Protein Fusion: out of frame {FGFR2:SRRT}	NA
EVT-P-0055975-T01-IM6-139	P-0055975-T01-IM6		FGFR2-STAT4	STAT4	in-frame	five_prime	123240312	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - STAT4 (NM_001243835) rearrangement: t(2;10)(q32.3;q26.13)(chr2:g.191936121::chr10:g.123240312) Protein Fusion: in frame {FGFR2:STAT4}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 5185-T01-IM6-141	P-0045 185-T01-IM6		FGFR2-TACC2	TACC2	in-frame	five_prime	123240014	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-479:FGFR2_c.8128-2249:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-003 1242-T01-IM6-142	P-0031 242-T01-IM6		FGFR2-TACC2	TACC2	unknown	five_prime	123239432	18	802	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2405:FGFR2_c.7896+237:TACC2inv Protein Fusion: mid-exon {FGFR2:TACC2}	NA
EVT-P-002 3364-T03-IM6-143	P-0023 364-T03-IM6		FGFR2-TACC2	TACC2	unknown	five_prime	123241587	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) rearrangement: c.2301+1625:FGFR2_c.34-5469:TACC2dup Antisense Fusion	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 4187-T01-I M6-144	P-0024 187-T0 1-IM6		FGFR2 -TACC 2	TAC C2	unknown	five_prime	123240 573	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) Fusion (FGFR2 exon 17 with TACC2 exon12) : c.2302-1038:FGFR2_c.8009:TACC2inv Protein Fusion: mid-exon {FGFR2:TACC2}	NA
EVT-P-005 0962-T01-I M6-145	P-0050 962-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-frame	five_prime	123241 844	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1368:FGFR2_c.7608+332:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-003 6358-T01-I M6-146	P-0036 358-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-frame	five_prime	123239 743	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2302-208:FGFR2_c.7344+468:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 7746-T01-I M6-149	P-0057 746-T0 1-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123241 998	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128 -402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-005 7746-T02-I M6-150	P-0057 746-T0 2-IM6		FGFR2 -TACC 2	TAC C2	in-fr ame	fiv e_ pri me	123241 998	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128 -402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	NA
EVT-P-002 2337-T06-I M6-151	P-0022 337-T0 6-IM6		FGFR2 -TBC1 D4	TBC 1D4	in-fr ame	fiv e_ pri me	123241 723	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2) (chr10:g.123241723::chr13: g.75946350) Protein Fusion: in frame {FGFR2:TBC1D4}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 2337-T04-IM6-152	P-0022 337-T04-IM6		FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946350) Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-000 5291-T01-IM5-153	P-0005 291-T01-IM5		FGFR2-TFEC	TFEC	unknown	five_prime	123241729	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26.13;q31.2)(chr10:g.123241729;chr7:g.115644112) Transcript fusion (FGFR2-TFEC)	NA
EVT-P-001 6640-T01-IM6-154	P-0016 640-T01-IM6		FGFR2-TTC28	TTC28	in-frame	five_prime	123241944	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement : t(10;22)(q26.11;q12.1)(chr10:g.123241944::chr22:g.28666001) Protein Fusion: in frame {FGFR2:TTC28}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051558-T01-IM6-155	P-0051558-T01-IM6		FGFR2-TTC28	TTC28	in-frame	five_prime	123243158	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1)(chr10:g.123243158::chr22:g.28605071) Protein Fusion: in frame {FGFR2:TTC28}	NA
EVT-P-0029726-T01-IM6-156	P-0029726-T01-IM6		FGFR2-UBP1	UBP1	in-frame	five_prime	123240062	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - UBP1 (NM_001128161) fusion: t(3;10)(p22.3;q26.13)(chr3:g.33449450::chr10:g.123240062) Protein Fusion: in frame {FGFR2:UBP1}	NA
EVT-P-0004089-T01-IM5-158	P-0004089-T01-IM5		FGFR2-VCL	VCL	unknown	five_prime	123242749	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2(NM_000141) - VCL (NM_014000) Rearrangement : c.2301+463:FGFR2_c.1032:VCLinv Protein fusion: mid-exon (FGFR2-VCL)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 9238-T02-I M7-159	P-0059 238-T0 2-IM7		FGFR2 -WAC	WAC	in-frame	five_prime	123242066	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+1146:FGFR2_c.1437+679:WACinv Protein Fusion: in frame {FGFR2:WAC}	NA
EVT-P-005 8031-T01-I M6-160	P-0058 031-T0 1-IM6		FGFR2 -WAC	WAC	in-frame	five_prime	123243180	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - WAC (NM_016628) fusion: c.2301+32:FGFR2_c.382-351:WACinv Protein Fusion: in frame {FGFR2:WAC}	NA
EVT-P-002 5807-T01-I M6-161	P-0025 807-T0 1-IM6		FGFR2 -YTHDF3	YTHDF3	unknown	three_prime	123242317	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain		YTHDF3 (NM_152758) - FGFR2 (NM_000141) rearrangement: t(8;10)(q12.3;q26.13)(chr8:g.64081967::chr10:g.123242317) Protein Fusion: mid-exon {YTHDF3:FGFR2}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-005 6399-T01-I M6-162	P-0056 399-T0 1-IM6		FGFR2 -ZFAT	ZFA T	in-fr ame	fiv e_ pri me	123240 282	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - ZFAT (NM_020863) rearrangement: t(8;10)(q24.2 2;q26.13)(chr8:g.135564146: :chr10:g.123240282) Protein Fusion: in frame {FGFR2:ZFAT}	NA
EVT-P-004 2241-T01-I M6-163	P-0042 241-T0 1-IM6		FGFR2 -ZMYM 4	ZM YM4	in-fr ame	fiv e_ pri me	123242 889	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - ZMYM4 (NM_005095) fusion: t(1;10)(p34.3;q26.13)(chr1:g.35833702::chr10:g.12 3242889) Protein Fusion: in frame {FGFR2:ZMYM4}	NA
EVT-P-001 6195-T01-I M6-164	P-0016 195-T0 1-IM6		GAB2- FGFR2	GA B2	in-fr ame	fiv e_ pri me	123242 812	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(1 0;11)(q26.11;q14.1)(chr10:g. 123242812::chr11:g.780305 53) Protein Fusion: in frame {FGFR2:GAB2}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-001 3756-T01-I M5-165	P-0013 756-T0 1-IM5		INA-F GFR2	INA	in-fr ame	thre e_ pr ime	123242 695	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		INA (NM_032727) - FGFR2 (NM_000141) Rearrangement : c.1066-350 0:INA_c.2301+517:FGFR2inv v Protein fusion: in frame (INA-FGFR2)	NA
EVT-P-001 3756-T03-I M6-166	P-0013 756-T0 3-IM6		INA-F GFR2	INA	in-fr ame	thre e_ pr ime	123242 695	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500: INA_c.2301+517:FGFR2inv Protein Fusion: in frame {INA:FGFR2}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_introductory_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-0002810-T01-IM3-169	P-0002810-T01-IM3		KIAA1217-FGFR2	KIAA1217	in-frame	five_prime	123240874	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			NA Protein fusion: in frame (FGFR2-KIAA1217)	NA
EVT-P-000386-T02-IM5-170	P-0000386-T02-IM5		MARVELD3-FGFR2	MARVELD3	unknown	five_prime	123243101	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13) (chr16:g.71660650::chr10:123243101) Antisense fusion	NA
EVT-P-0024094-T01-IM6-172	P-0024094-T01-IM6		MYH9-FGFR2	MYH9	in-frame	five_prime	123240418	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240418::chr22:g.36682296) Protein Fusion: in frame {FGFR2:MYH9}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003 2208-T01-IM6-173	P-0032 208-T01-IM6		MYO18B-FGFR2	MYO18B	in-frame	five_prime	123241929	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929::chr22:g.26280218) Protein Fusion: in frame {FGFR2:MYO18B}	NA
EVT-P-002 1975-T01-IM6-175	P-0021 975-T01-IM6		NID2-FGFR2	NID2	out-of-frame	three_prime	123240637	18	768	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain		NID2 (NM_007361) - FGFR2 (NM_000141) rearrangement: t(10;14)(q26.13;q22.1)(chr10:g.123240637::chr14:g.52482743) Protein Fusion: out of frame {NID2:FGFR2}	NA
EVT-P-001 2423-T01-IM5-176	P-0012 423-T01-IM5		NOL4-FGFR2	NOL4	in-frame	five_prime	123242054	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain ; Immunoglobulin I-set domain			FGFR2 (NM_000141) - NOL4 (NM_003787) Rearrangement :t(10;18)(q26.13;q12.1)(chr10:g.123242054::chr18:g.31563418) Protein fusion: in frame (FGFR2-NOL4)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023103-T01-IM6-177	P-0023103-T01-IM6		NOL4-FGFR2	NOL4	in-frame	five_prime	123241921	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused to NOL4 exons 10-11): t(10;18)(q26.13;q12.1)(chr10:g.123241921::chr18:g.31518701) Protein Fusion: in frame {FGFR2:NOL4}	NA
EVT-P-0008693-T01-IM5-178	P-0008693-T01-IM5		NRAP-FGFR2	NRAP	in-frame	five_prime	123242277	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - NRAP (NM_198060) rearrangement: c.2301+935:FGFR2_c.2458-427:NRAPdel Protein fusion: in frame (FGFR2-NRAP)	NA
EVT-P-0006328-T01-IM5-179	P-0006328-T01-IM5		NRBF2-FGFR2	NRBF2	in-frame	five_prime	123241245	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - NRBF2 (NM_030759) rearrangement: c.2302-1710:FGFR2_c.157-364:NRBF2inv Protein fusion: in frame (FGFR2-NRBF2)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004 9071-T01-IM6-180	P-0049 071-T01-IM6		NRG3-FGFR2	NRG3	out-of-frame	thre_prime	123244882	16	732	True	disrupted	0.09386281588447654	True		Immunoglobulin domain	Protein tyrosine and serine/threonine kinase	NRG3 (NM_001010848) - FGFR2 (NM_000141) rearrangement: c.823+108972:NRG3_c.2195+27:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-002 2337-T01-IM6-181	P-0022 337-T01-IM6		NRG3-FGFR2	NRG3	out-of-frame	thre_prime	123241723	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin domain		NRG3 (NM_001010848) - FGFR2 (NM_000141) Rearrangement : c.953+89877:NRG3_c.2301+1489:FGFR2inv Protein Fusion: out of frame {NRG3:FGFR2}	NA
EVT-P-003 2171-T03-IM6-182	P-0032 171-T03-IM6		OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin domain; Immunoglobulin domain			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN}	NA

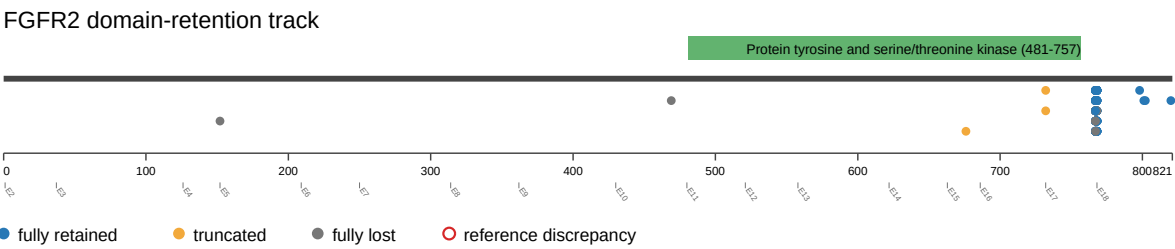
event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_ etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-003 2171-T01-I M6-183	P-0032 171-T0 1-IM6		OPTN- FGFR2	OPT N	in-fr ame	fiv e_ pri me	123239 789	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254: FGFR2_c.553-1543:OPTNin v Protein Fusion: in frame {FGFR2:OPTN}	NA
EVT-P-002 2781-T01-I M6-184	P-0022 781-T0 1-IM6		PDZR N4-FG FR2	PDZ RN4	in-fr ame	fiv e_ pri me	123241 997	17	767	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1 ;q26.1)(chr6:g.130408449::c hr10:g.123239680) Protein Fusion: in frame {FGFR2:PDZRN4}	NA
EVT-P-004 4768-T01-I M6-187	P-0044 768-T0 1-IM6		RNF14 5-FGF R2	RNF 145	out-fr ame	thre e_ pr ime	123243 127	17	767	True	lost	0.0	False		Prote in tyr osine and serin e/thr eoni ne ki nase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		RNF145 (NM_144726) - FGFR2 (NM_000141) rearrangement: t(5;10)(q33.3 ;q26.13)(chr5:g.158598146:: chr10:g.123243127) Protein Fusion: out of frame {RNF145:FGFR2}	NA

event_id	sampl e_id	pa tie nt _i d	fusion _name	part ner _ge ne	fra me _st atu s	tar get _r ole	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intro nic_bre akpoint	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 6235-T01-I M5-188	P-0006 235-T0 1-IM5		ROCK 1-FGFR R2	RO CK1	in-fr ame	fiv e_ pri me	123240 133	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - ROCK1 (NM_005406) rearrangement: t(10;18)(q26. 13;q11.1)(chr10:g.12324013 3::chr18:g18680051) Protein fusion: in frame (FGFR2-ROCK1)	NA
EVT-P-003 9370-T01-I M6-189	P-0039 370-T0 1-IM6		RPAP3 -FGFR 2	RPA P3	in-fr ame	thr ee_ pr ime	123242 832	17	767	True	lost	0.0	False		Prote in tyr osine and se rine/thr eonine kinase ; Im mun oglo bulin dom ain; I mmu nogl obuli n I-set dom ain		RPAP3 (NM_024604) - FGFR2 (NM_000141) rearrangement: t(10;12)(q26. 13;q13.11)(chr10:g.1232428 32::chr12:g.48092574) Protein Fusion: in frame {RPAP3:FGFR2}	NA
EVT-P-002 5074-T01-I M6-190	P-0025 074-T0 1-IM6		SHTN1 -FGFR 2	SHT N1	in-fr ame	fiv e_ pri me	123240 122	18	768	True	retai ned	1.0	False	Protei n tyros ine and se rine/thr eonine kinase ; Immu noglob ulin do main; I mmun oglobu lin I-set d omain			FGFR2 (NM_000141) - KIAA1598 (NM_001127211) Fusion (FGFR2 exon 17 with KIAA1598 exon 5): c.2302-5 87:FGFR2_c.267+2596:KIA A1598 Protein Fusion: in frame {FGFR2:KIAA1598}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_2_effect_on_frame
EVT-P-003 2507-T01-I M6-191	P-0032 507-T0 1-IM6		SORBS1-FGFR2	SORBS1	unknown	five_prime	123242268	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) Rearrangement : c.2301+944:FGFR2_chr10:g.97250518del Transcript Fusion {FGFR2:SORBS1}	NA
EVT-P-002 2337-T01-I M6-192	P-0022 337-T0 1-IM6		TBC1D4-FGFR2	TBC1D4	in-frame	five_prime	123241723	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain			FGFR2 (NM_000141) - TBC1D4 (NM_014832) Rearrangement : t(10;13)(q26.13;q22.2)(chr10:g.123241723::chr13:g.75946348) Protein Fusion: in frame {FGFR2:TBC1D4}	NA
EVT-P-002 1788-T01-I M6-193	P-0021 788-T0 1-IM6		VPS26A-FGFR2	VPS26A	in-frame	three_prime	123242226	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase ; Immunoglobulin domain; Immunoglobulin I-set domain		VPS26A (NM_004896) - FGFR2 (NM_000141) Rearrangement : c.4-1035:VPS26A_c.2301+986:FGFR2inv Protein Fusion: in frame {VPS26A:FGFR2}	NA

Figures

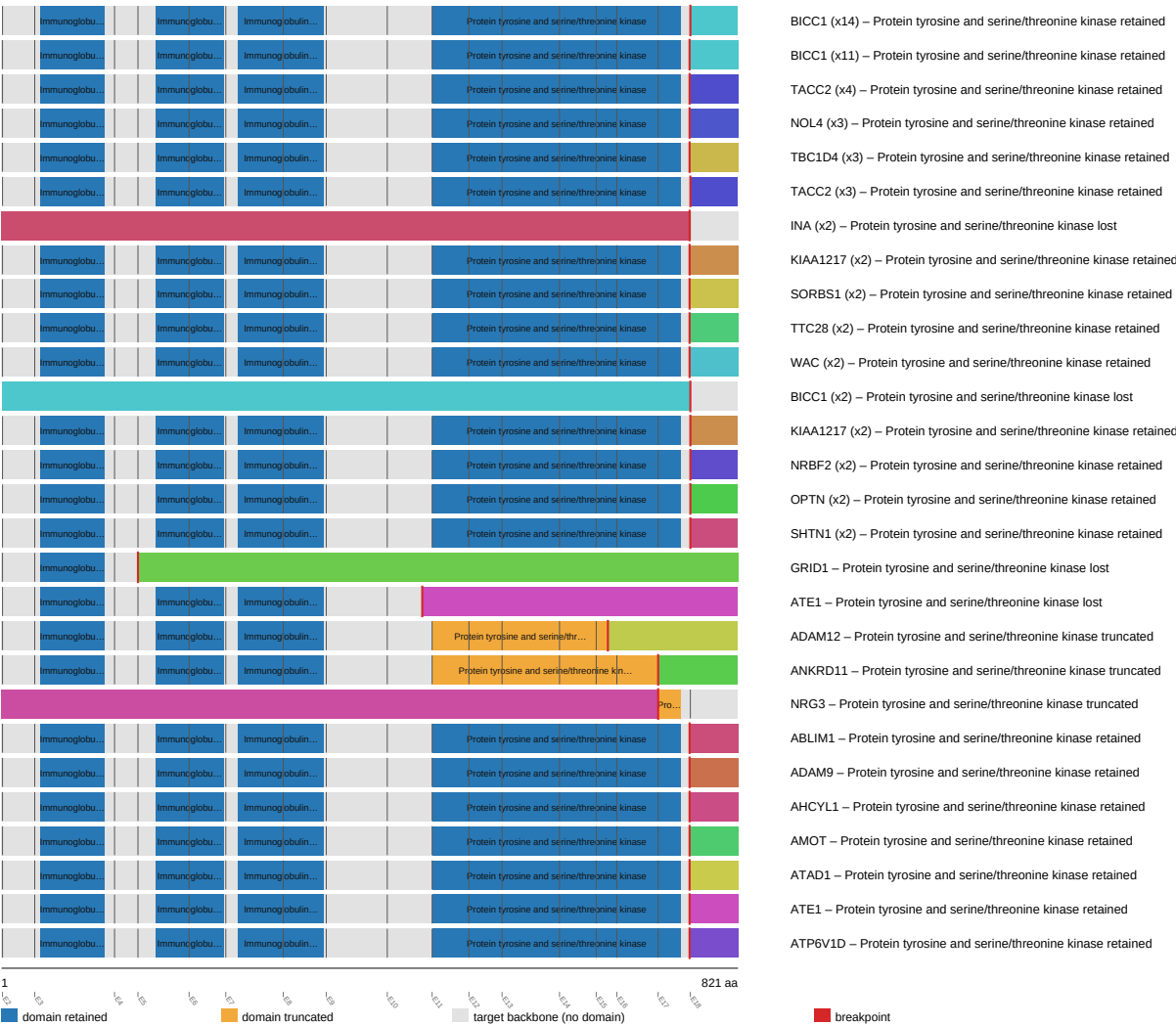
domain retention outliers



fusion schematic

FGFR2 fusion-transcript schematic

partner block → breakpoint → retained FGFR2 portion (domain-colored, exon ticks)



Showing the top 28 of 92 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026

